

Comparative Analysis of Sentiment Analysis on X, formerly known as Twitter, Data between BERT and RoBERTa

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Abstract—This project explores sentiment analysis as an application of artificial intelligence, focusing on detecting and interpreting the meaning, intent, and tone behind textual data. We studied and compared two modern AI models, BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa (A Robustly Optimized BERT Pretraining Approach), to assess their ability to classify text as positive, negative, or neutral, and to detect emotions and intent. The comparison is used to determine which model could process real-world data (e.g., social media posts, reviews, or feedback) and provide insights into user attitudes and opinions. This approach has valuable applications in fields such as customer service, marketing, and public opinion analysis, where understanding sentiment is critical to decision-making.

Keywords— *Sentiment Analysis, BERT, RoBERTa, text data, emotions, intent*

I. INTRODUCTION

With the rise of generative pre-trained transformers and the vast text data available from X, we considered creating a model that could serve as a tool to help detect users' or customers' emotions and the intent behind their posts. We'll use two powerful language representation models: BERT (Bidirectional Encoder Representations from Transformers), which was introduced by Devlin [1], and RoBERTa (A Robustly Optimized BERT Pretraining Approach) presented by Liu [2]. BERT was the first model in NLP (natural language processing) to obtain an understanding of text using bidirectional transformers. Older models would read text in sequence, either in a left-to-right or a right-to-left pattern. BERT is the first model to read text fully bidirectionally. While BERT came from Google, RoBERTa has its beginnings in Facebook (FB) AI. The creation of RoBERTa was due to Liu's belief that it was undertrained, and the

researchers provided an improved version of BERT, which they called RoBERTa. Both models only use the encoder part of the Transformer architecture (mentioned later in the paper).

BERT has been pretrained with BooksCorpus and English Wikipedia; it was created for easy application with only an additional output layer to create models for different tasks (as in the case of RoBERTa) and uses an encoder-only architecture. "BERT is the first fine-tuning-based representation model that achieves state-of-the-art performance on a large suite of sentence-level and token-level tasks, outperforming many task-specific architectures." [1]

BERT can capture inner contextual relationships among the words in a text sequence bidirectionally, opening a whole range of advanced NLP applications like text classification, sentiment analysis, and question answering.

The FB team that created RoBERTa wanted to replicate the BERT study, carefully measuring many key hyperparameters and the training data size. Their goal was to showcase that BERT could match or exceed the performance of models after it; the key was that it was undertrained. They showed in their paper that hyperparameters have a significant impact on the results [2].

II. LITERATURE REVIEW

Recent studies have paid close attention to how BERT and RoBERTa perform side by side in sentiment analysis. Tan et al. (2023) explored a hybrid RoBERTa-GRU model for X sentiment classification and found that RoBERTa reached 90.67% accuracy, slightly higher than BERT's 89.36% [3], suggesting that combining contextual embeddings with recurrent layers can sharpen performance. In contrast, Joshy and Sundar (2022) tested both models on COVID-19-related tweets and reported that BERT (90.43%) clearly outperformed

RoBERTa (76.03%) [4], showing that domain-specific data still plays a big role in model effectiveness. Li et al. (2023) examined airline customer reviews and found RoBERTa (96.97%) edged out BERT (95.32%) [5], likely because of RoBERTa's deeper pre-training and better handling of subtle sentiment cues. Rahman et al. (2024) compared the two across multiple datasets—including IMDb, X, and Sentiment140—and observed that RoBERTa consistently came out ahead, with accuracies of 92.36%, 80.74%, and 82.25%, respectively [6]. Meanwhile, Chatzimina et al. (2024) studied Greek clinical dialogues and found the opposite trend: BERT (95.48%) outperformed RoBERTa (91.43%) [7], suggesting that the model's success can depend heavily on the language and domain of the data.

III. METHODOLOGY

This study aims to compare the performance of BERT and RoBERTa for the task of X sentiment analysis. The objective is to evaluate how differences in pretraining strategy, data size, and optimization affect each model's ability to correctly classify the sentiment polarity of tweets (positive, negative, or neutral). Both models share the same underlying Transformer encoder architecture but vary in their training configurations and data preprocessing.

Misspellings and irregularities in words can occur when messages are written on X, Hu et al. (2020) discussed how BERT is capable of effectively correcting spelling errors. BERT can predict a masked word that would make sense in each context [8]. A misspelled word can be masked in BERT to allow the model to change it with a word that is spelled correctly.

Emojis are part of how humans express themselves in this digital age, so when text data, including emojis, is fed to BERT, how accurate will the sentiment analysis be? Khan et al. (2025) published their findings earlier this year. For their proposed approach with Variant II, emojis were substituted with corresponding tag words from an inbuilt emoji dictionary. They showed the experiment's results in Table 2 of their paper, with BERT (Variant II) having a 91% accuracy [9].

A. Transformer Architecture

The Transformer architecture, introduced by Vaswani et al. (2017), is a neural network model that processes sequential data using an attention-based mechanism rather than recurrent neural networks (RNN) or convolutional neural networks (CNN). Its key innovation, the self-attention mechanism [10], allows the model to compute contextual relationships between all words in a sentence simultaneously, capturing both local and long-range dependencies efficiently. This parallelizable structure enables faster training and more robust representation learning compared to RNN or CNN-based models [11].

Most modern neural sequence-to-sequence models are built upon an encoder-decoder architecture. In this framework, the encoder transforms an input sequence of symbol embeddings ($x_1, \dots, x_n$) into a sequence of continuous vector representations ($z_1, \dots, z_n$). The decoder then uses these representations to produce an output sequence ($y_1, \dots, y_m$), generating one element

at a time. The process is auto regressive, meaning that the decoder conditions each new output on the tokens it has already produced. The Transformer model adopts this same general structure but replaces recurrent operations with layers composed of self-attention mechanisms and position-wise feed-forward networks in both the encoder and decoder components [11], as illustrated in Figure 1.

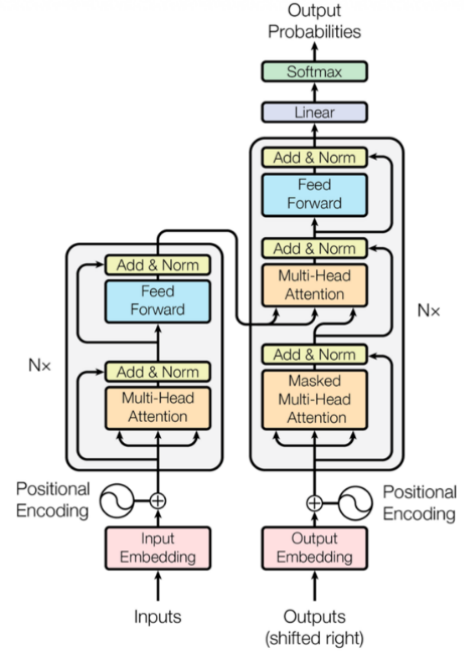


Figure 1. The Transformer - model architecture. [11]

In the context of BERT and RoBERTa, only the encoder part of the Transformer is used, since both models are designed for language understanding rather than sequence generation.

B. Encoder and Decoder Stacks

Encoder:

The encoder consists of a stack of six identical layers ($N = 6$). Each layer is made up of two main sublayers. The first sublayer implements a multi-head self-attention mechanism, which allows the model to attend to different positions within the same input sequence simultaneously. The second sublayer is a position-wise fully connected feed-forward network, applied independently to each position.

To stabilize training and preserve information flow, a residual connection is applied around each sublayer, followed by layer normalization. In other words, the output of each sublayer is computed as: $\text{LayerNorm}(x + \text{Sublayer}(x))$, where $\text{Sublayer}(x)$ represents the transformation performed by the sublayer. To ensure compatibility across residual paths, all sublayers and embedding layers produce outputs with a consistent dimensionality of $d_{\text{model}} = 512$. [11]

Decoder:

The decoder also consists of six identical layers ($N = 6$). Each layer includes the same two sublayers found in the

encoder, but with an additional third sublayer that performs multi-head attention over the encoder’s output representations. Like the encoder, each sublayer is wrapped with a residual connection and followed by layer normalization.

Furthermore, the decoder’s self-attention sublayer is modified through masking to prevent any position from attending to future tokens in the output sequence. This design ensures that the prediction for position i depends only on previously generated outputs (positions less than i). The combination of masking and shifted output embeddings enforces the model’s auto-regressive nature during generation. [11]

C. BERT:

BERT mainly consists of stacked transformer encoders, which were introduced by Vaswani et al. [11]. Transformer models have several self-attention heads within each encoder layer, allowing the model to allocate attention to different aspects of a sentence simultaneously.

BERT comes in two primary configurations:

- BERT Base: 12 layers, 12 attention heads, 110 million parameters
- BERT Large: 24 layers, 16 attention heads, 340 million parameters

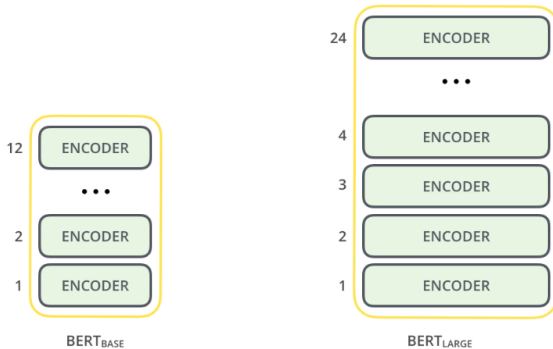


Figure 2. Architectural comparison between BERT Base and BERT Large

These configurations were intended to ensure optimal trade-offs between performance and handling power. In this case, the large model is especially recommended for tasks that are resource-intensive [1].

D. RoBERTa

RoBERTa is an improved variant of BERT designed to address limitations in the original pre-training procedure. While RoBERTa retains the same transformer architecture as BERT, it achieves significantly better performance by modifying the training strategy rather than changing the model design. Specifically, RoBERTa removes the Next Sentence Prediction (NSP) objective, increases the size of the training corpus, and trains for substantially more steps using larger batch sizes. It

also employs dynamic masking instead of BERT’s static masking, allowing the model to see different masked versions of the same sentence across training epochs. These changes result in a more robust language representation that generally outperforms BERT on a wide range of downstream NLP tasks [2].

E. Fine-Tuning and Adaptability

After pre-training, BERT can be further trained on labeled data for specific tasks by customizing the top layers of the encoder. The reason this step is swift is that the pre-trained weights have already been useful for different tasks in NLP.

As consolidated in the article by Sun et al [12]. “How to Fine-Tune BERT for Text Classification,” the most impactful tuning relates to the learning rate, batch size (as mentioned in section D. RoBERTa), and number of epochs. They recommend learning rates in the $2e-5$ to $5e-5$ range, and shortened overall time to train in order to avoid overfitting.

This flexibility makes BERT effective for diverse downstream applications such as:

- Sentiment analysis
- Named entity recognition
- Text classification
- Question answering

IV. EXPERIMENT

A. Dataset Preprocessing

For the sentiment classification task, we used a dataset of 74,682 total entries for training and 1,000 entries for validation. During model development and evaluation, we trained exclusively on the 74k-entry training set and used the 1k-entry validation set to monitor performance and prevent overfitting.

To process the raw text, we mapped all sentiment labels to integer class IDs (e.g., negative, neutral, positive). This mapping was shared across all models to ensure consistent evaluation. We then tokenized the input tweets using the corresponding pretrained tokenizer of each model—BERT-base-uncased and RoBERTa-base—with a maximum sequence length of 128 tokens. Padding and truncation were applied to maintain uniform input size.

B. Fine-Tuning Setup

Both models were fine-tuned using the AdamW optimizer, which is widely used for transformer-based architectures because it helps maintain stable training and handles weight decay effectively. Following common recommendations in transformer literature, we set the learning rate to $2e-5$, which offers a balance between convergence speed and avoiding overshooting minima during fine-tuning. We trained both models for 3 epochs, a standard choice for adapting pretrained language models to downstream text-classification tasks.

To handle the large training set, we used model-specific batch sizes based on GPU memory constraints: both BERT and

RoBERTa were trained with a batch size of 16. For evaluation, the same batch sizes were kept to ensure consistency in loss and accuracy reporting. During training, we tracked both loss and accuracy across epochs for the training and validation sets. After fine-tuning, we saved each model and tokenizer for reproducibility and further analysis.

Finally, model performance was assessed on the validation set using standard metrics: accuracy, precision, recall, F1-score, and a confusion matrix. These metrics allowed us to evaluate how well each model handled the multi-class sentiment classification task and to compare the behavior of BERT and RoBERTa under similar fine-tuning configurations.

C. Learning rate sensitivity analysis

Although both BERT and RoBERTa were ultimately fine-tuned using a learning rate of $2e-5$, we conducted an additional sensitivity analysis to understand how different learning rates influence model performance. Each model was trained for three epochs on the same Twitter/X dataset using five learning rates: $4e-4$, $2e-5$, $4e-5$, $4e-6$, and $4e-7$. For each setting, we recorded the final validation accuracy from the last epoch. The following figures show how performance varies as a function of the learning rate.

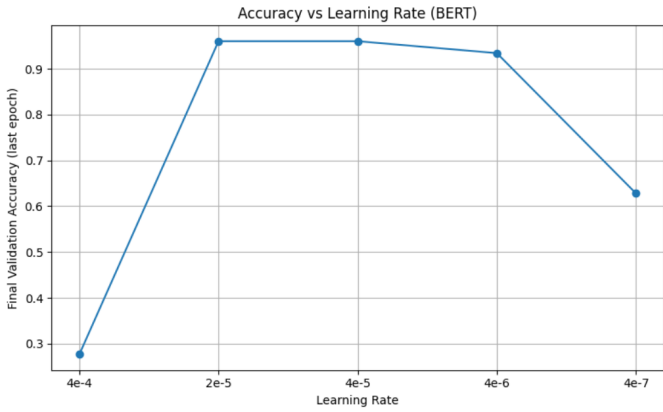


Figure 3. Effect of learning rate on BERT’s final validation accuracy after 3 epochs.

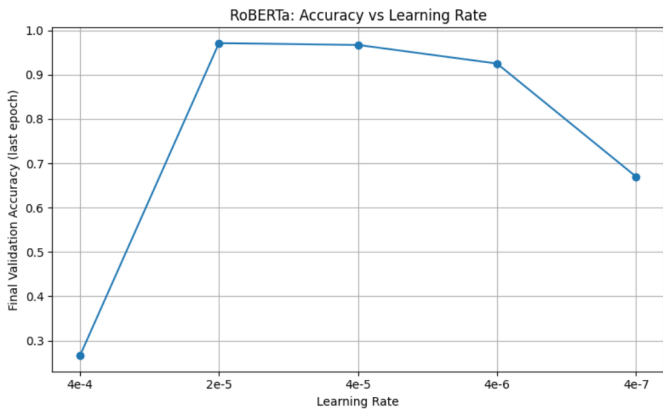


Figure 4. Effect of learning rate on RoBERTa’s final validation accuracy after 3 epochs.

These results confirm established findings in the literature. Sun et al. [12] noted that transformer-based models fine-tune best within the $2e-5$ to $5e-5$ range, which is consistent with our observations. Both BERT and RoBERTa achieved their strongest performance at $2e-5$, validating our choice for the final experiments.

Interestingly, RoBERTa showed a sharp, yet slightly better, stability at smaller learning rates, whereas BERT experienced a sharper drop-off. This aligns with RoBERTa’s more aggressive pretraining strategy, which often yields more robust downstream fine-tuning.

V. RESULTS

Results for the sentiment classification task are presented in the following tables as accuracy, F1 scores, and output loss values for both BERT and RoBERTa models. These metrics were computed on the held-out validation set of 1,000 samples. Lower loss values indicate better model convergence. Our findings from the development phase are shown in the tables, and the final F1-score comparison is summarized in the following tables

Table 1. Classification Results Across Emotion Categories for BERT and RoBERTa

Emotion / Model	Irrelevant	Negative	Neutral	Positive
BERT	157/172	256/266	280/285	267/277
RoBERTa	162/172	259/266	280/285	270/277

Table 1. Validation Output Loss for BERT and RoBERTa

Model	Output Loss
BERT	0.1643
RoBERTa	0.1060

Table 2. F1 Scores for BERT and RoBERTa

Model	F1 Score
BERT	0.96
RoBERTa	0.97

The BERT model reached a validation accuracy of 0.96, with strong performance across all three emotion classes. Its classification report shows high precision and recall, with F1 scores ranging from 0.95 to 0.97. The confusion matrix indicates

that BERT occasionally misclassifies Irrelevant tweets but performs consistently well on Negative and Positive categories.

During training, BERT showed steady convergence. Training loss decreased from 0.75 in the first epoch to 0.14 in the third, while validation accuracy rose from 0.94 to 0.96. The test loss curve reached its lowest point at epoch 2, suggesting that the optimal point before slight overfitting occurs is around the second epoch.

RoBERTa achieved a stronger overall performance, with a validation accuracy of 0.971, outperforming BERT across all major metrics. Its F1 scores ranged from 0.96 to 0.98, with particularly high recall for the Neutral and Negative classes. The confusion matrix shows fewer misclassifications than BERT, especially for Positive tweets.

Loss values also confirm RoBERTa's advantage: validation loss dropped sharply from 0.416 in epoch 1 to 0.106 in epoch 3. Training accuracy increased steadily across epochs, reaching 0.91 by the final epoch.

Overall, both models performed well on this multi-class X sentiment task. BERT demonstrated strong baseline performance with high accuracy and balanced precision-recall values, but RoBERTa consistently delivered better results across all evaluation metrics. RoBERTa achieved:

- higher validation accuracy (0.971 vs. 0.96),
- lower validation loss (0.106 vs. 0.164),
- stronger F1 scores across all classes, and
- fewer misclassifications in the confusion matrix.

These results align with expected behavior, as RoBERTa's training strategy and dynamic masking generally provide stronger downstream performance on text-classification tasks.

VI. CONCLUSIONS

The advent of BERT was a significant moment in the history of NLP. It implemented a new method of attention in a bidirectional way, incorporated other transformer architecture elements, and innovated the pre-training and fine-tuning model training stages, which set a solid initial groundwork for following models and architectures like RoBERTa.

The reviewed papers collectively demonstrate:

- BERT's architectural innovation
- Its interpretability and visualization
- Its fine-tuning techniques
- Its applications in sentiment analysis
- And its theoretical insights

The main theoretical idea in this project is that BERT and RoBERTa, even though they share the same transformer structure, perform differently because they were trained in different ways. By importing both pretrained models and running them on the same Twitter/X dataset, we can see how changes in their training strategies—like RoBERTa using more data and removing the Next Sentence Prediction step—affect their ability to understand sentiment. This helps show that stronger or more optimized pretraining can lead to better results, even when the models themselves look very similar.

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